

FORMAN CHRISTAIN COLLEGE

(A Chartered University)

Synopsis for MS Degree in Computer Science

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1. Title: Integrating Smartphone-Derived Trigger Data with Interictal EEG for Enhanced Epilepsy Classification and Prediction

2. Introduction

Epilepsy is a chronic neurological disorder that affects over 50 million people worldwide. Accurate diagnosis and early prediction are critical for improving patient outcomes, but these remain difficult challenges. The gold-standard diagnostic tool, electroencephalography (EEG), clearly captures ictal activity during seizures but often appears visually normal during interictal periods, even in epileptic patients. This makes it difficult to distinguish interictal EEGs from the EEGs of healthy individuals, leading to delays in diagnosis and uncertainty in treatment.

At the same time, neurological research shows that seizures are frequently triggered by lifestyle, physiological, and environmental factors such as poor sleep, stress, hormonal fluctuations, and environmental conditions. With smartphones and wearable devices becoming widespread, many of these triggers can now be measured continuously using built-in sensors, APIs (e.g., weather, pollution), and patient diaries.

This project proposes to combine **EEG biomarkers** with **smartphone-derived trigger data** to design a novel multimodal machine learning framework. The goal is to enhance classification of epileptic patients versus healthy controls (even when EEGs look normal) and provide predictive insights about seizure risk.

3. Research Problem

EEG-only models struggle to detect subtle, hidden patterns in interictal recordings, while trigger-only data lacks direct neurophysiological evidence. Integrating these two complementary streams could allow us to (1) identify interictal biomarkers in EEG that correlate with seizure triggers, and (2) develop context-aware prediction models that outperform EEG-only baselines.

Research Question:

Can multimodal integration of interictal EEG and smartphone-derived seizure trigger data improve epilepsy classification and provide reliable seizure prediction compared to EEG-only approaches?

4. Objectives

1. Develop a multimodal deep learning framework for integrating EEG and trigger data.
2. Extract subtle interictal biomarkers using deep and hybrid models.
3. Incorporate lifestyle/environmental triggers into predictive modeling.
4. Compare multimodal models with unimodal (EEG-only and trigger-only) baselines.
5. Demonstrate feasibility of smartphone-supported epilepsy monitoring.

5. Proposed Methodology

Data Sources

- **EEG Data:** Public datasets (e.g., TUH EEG Corpus, CHB-MIT Scalp EEG Database, EPILEPSIAE) that provide ictal, interictal, and healthy control recordings. Only interictal segments will be used for epilepsy patient data.
- **Trigger Data:** Smartphone-collected data simulated or obtained via pilot studies, including sleep quality, stress (HRV from camera PPG), physical activity, environmental conditions (from APIs), and patient diary inputs.

Step 1 – EEG Preprocessing & Feature Engineering

- Preprocess raw EEG signals (artifact removal, band-pass filtering, ICA).
- Extract handcrafted features (time-domain, frequency-domain, entropy, connectivity).
- Train deep models (CNNs, Transformers, Graph Neural Networks) to learn latent embeddings from raw EEG segments.

Step 2 – Trigger Data Processing

- Collect smartphone sensor and diary data.
- Engineer features for sleep patterns, stress levels, motion activity, environmental exposure.
- Encode temporal patterns using LSTMs or Transformers to capture time dependencies (e.g., cumulative sleep deficit).

Step 3 – Multimodal Fusion

- **Early Fusion:** Concatenate EEG embeddings and trigger embeddings into a joint feature space.
- **Late Fusion:** Train separate models for EEG and triggers, then combine outputs using ensemble methods.
- **Hybrid Fusion with Attention:** Assign adaptive weights to EEG vs. trigger signals depending on context (e.g., stress-related triggers may be more predictive for some patients).

Step 4 – Classification & Prediction

- **Classification Task:** Distinguish between interictal epileptic EEGs and healthy control EEGs.
- **Prediction Task:** Estimate seizure likelihood by aligning trigger history with interictal EEG.
- Use evaluation metrics such as accuracy, F1-score, AUROC for classification, and calibration/error measures for prediction reliability.

Step 5 – Validation

- Perform cross-validation across datasets and sites to test generalizability.
- Conduct ablation studies to measure the contribution of trigger data vs. EEG.
- Analyze explainability using saliency maps and feature attribution to identify relevant EEG biomarkers and trigger factors.

6. Expected Contributions

- A novel multimodal machine learning framework combining EEG and smartphone-derived triggers.
- Identification of hidden interictal EEG biomarkers contextualized by lifestyle and environmental conditions.
- Demonstration of how digital phenotyping can complement traditional neurophysiology.
- A proof-of-concept system showing the feasibility of smartphone-based epilepsy monitoring and prediction.

7. References (Recent Works)

1. Rudroff, T. (2025). Digital Biomarkers and AI for Remote Monitoring of Fatigue Progression in Neurological Disorders: Bridging Mechanisms to Clinical Applications. *Brain Sciences*, 15(5), 533. <https://doi.org/10.3390/brainsci15050533>

2. Abadal, S., Galván, P., Mármol, A., Mammone, N., Ieracitano, C., Lo Giudice, M., Salvini, A., & Morabito, F. C. (2024). Graph neural networks for electroencephalogram analysis: Alzheimer’s disease and epilepsy use cases. *Neural Networks*, 175, 106792. <https://doi.org/10.1016/j.neunet.2024.106792>

3. Xiao, T., Wang, Z., Zhang, Y., Lv, H., Wang, S., Feng, H., & Zhao, Y. (2023). Self-supervised Learning with Attention Mechanism for EEG-based seizure detection. *Biomedical Signal Processing and Control*, 87, 105464. <https://doi.org/10.1016/j.bspc.2023.105464>

8. Graphical Representation:

